Lesson 10.6 - P-Value Method for Hypothesis Testing

> **Video:** P-Value Method For Hypothesis Testing by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

Two ways to make the decision

	Traditional (critical-value) method	**P-value method**
Compare	calculated z **vs** critical z	**p-value vs** α
Reject H_0 when	z is in the rejection region	p-value $< \alpha$
Fail to reject when	z is in the fail-to-reject region	p-value $\geq \alpha$

Significance level: $\alpha = 1 - \text{confidence level}$.

p-value = the area in the tail(s) beyond the calculated test statistic:

Test	p-value
Left-tailed	area to the **left** of z
Right-tailed	area to the **right** of z
Two-tailed	**2 ×** the area in the tail beyond $\lvert z \rvert$

Example 1 - two-tailed (town weights)

Town XYZ's mean weight is claimed to be **168 lb**. A nutritionist thinks it is different. A sample of **36** people has $\bar{x} = 169.5$ lb and $s = 3.9$ lb.

(a)

$$H_0 : \mu = 168 \qquad H_a : \mu \neq 168$$

(b) Test at 95% confidence ($\alpha = 0.05$), two-tailed, $n \geq 30$ so use z:

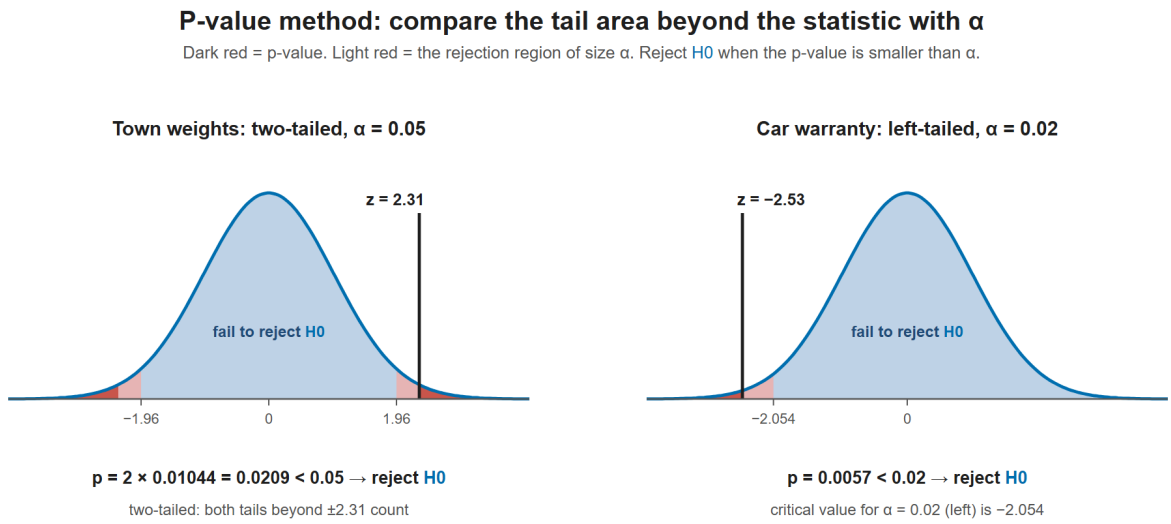
$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{169.5 - 168}{3.9/\sqrt{36}} = \frac{1.5}{0.65} \approx 2.31$$

Traditional method: the critical values are ± 1.96 (left area 0.975). $2.31 > 1.96$, so it is in the rejection region → **reject** H_0.

P-value method:

- The table gives a left area of 0.98956 for $z = 2.31$.
- Right tail: $P(z > 2.31) = 1 - P(z < 2.31) = 1 - 0.98956 = 0.01044$.
- Two-tailed p-value: $2 \times P(z > |z_c|) = 2 \times 0.01044 = 0.02088$.

$0.02088 < 0.05$ → **reject** H_0. At the 5% level there is enough evidence that the mean weight is not 168 lb.



The p-value (dark red) is the tail area beyond the test statistic. For a two-tailed test, both tails count. In both examples the p-value is smaller than α (the light red region), so H_0 is rejected.

Example 2 - left-tailed (car warranty)

A factory gives a **5-year** warranty on the engine and transmission. An engineer believes failures happen **in less than 5 years**. A sample of **40** cars has $\bar{x} = 4.8$ years and $s = 0.5$ years.

(a)

$$H_0 : \mu \geq 5 \qquad H_a : \mu < 5$$

(b) Is there evidence at the 2% significance level ($\alpha = 0.02$, 98% confidence) to revise the warranty?

This is left-tailed (shade the left only), and $n = 40 \geq 30$ so use z:

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{4.8 - 5}{0.5/\sqrt{40}} = \frac{-0.2}{0.07906} \approx -2.53$$

The p-value is the area to the left of −2.53 = **0.0057**.

$0.0057 < 0.02$ → **reject** H_0. There is enough evidence that the mean time to failure is under 5 years, so the warranty should be revised.

Key takeaways

- p-value = the probability, assuming H_0 is true, of a result at least as extreme as the one observed.
- **Small p-value (< α) → reject H_0.** Large p-value (≥ α) → fail to reject.
- For a two-tailed test, **double** the one-tail area.
- The p-value method and the critical-value method always give the same decision.